

openKylin 3.0 offers deeper integration with AI agents

Open source operating system project openKylin has released version 3.0, introducing deeper artificial intelligence (AI)-agent integration while expanding its open Linux ecosystem across developers, hardware and computing environments.

The release brings KylinBot into the OS layer, allowing it to understand voice and text requests, retain context across interactions, and perform system tasks. Integration with Kylin-CUA, a computer-use automation tool, gives KylinBot access to functions such as system settings and device management.

Rather than limiting AI capabilities to its own agent, openKylin also supports third-party tools including OpenClaw and WorkBuddy, allowing developers to combine AI capabilities for different use cases. The project says its open ecosystem and large-scale AI resources can lower barriers to developing and using AI agents.

“The shift from ‘AI as an application’ to ‘AI as an operating system’ is not simply a feature upgrade, but a transformation of the underlying architecture,” said Wu Qingbo, chair of the openKylin technical committee. “An open ecosystem will play an important role in this process, allowing developers to build and extend intelligent capabilities in a more transparent and secure way while adapting to a growing hardware ecosystem.”

openKylin 3.0 also upgrades to the Linux 7.0 kernel, optimises more than 180 core components, expands reduced instruction set computer five (RISC-V) support and increases Rust use in system tools. The project targets AI-powered devices, robotics, industrial automation, high-performance computing and cloud computing.

The architecture delivers up to 40 trillion operations per second (TOPS) of AI performance, targeting workloads such as computer vision, machine learning, local large language models and agentic AI applications running directly at the edge.

However, the board’s key electronics feature is its ‘dual-brain’ architecture. Alongside the Qualcomm processor sits an STMicroelectronics STM32H5 microcontroller dedicated to real-time workloads. This separation allows the high-performance processor to manage AI inference and operating system tasks while the microcontroller handles deterministic operations such as sensing, actuation and motor control.



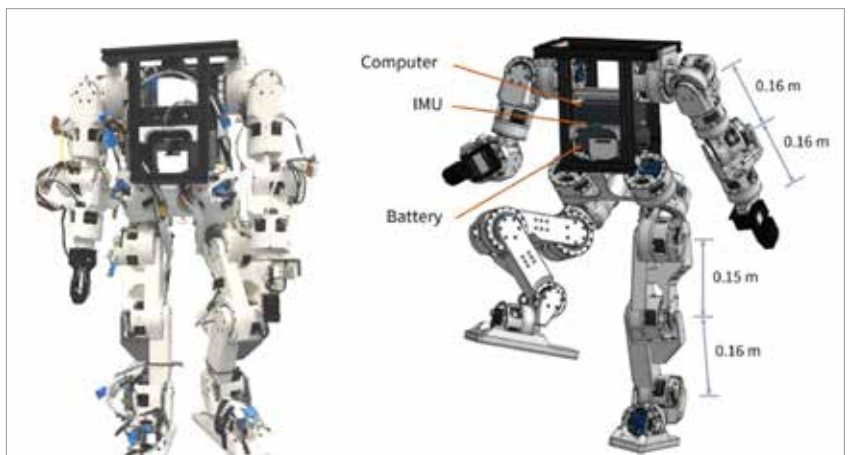
The board includes 16GB of LPDDR5 RAM and 64GB of eMMC storage, with an onboard M.2 slot supporting NVMe SSD expansion. Unlike the compact microcontroller boards traditionally

associated with Arduino, the board is physically larger and requires active cooling with a heatsink and fan to sustain demanding workloads. The platform is also intended to bridge the gap between prototyping and commercial deployment. Through Arduino’s ‘Works with Arduino’ programme, developers can potentially move applications developed on the board towards production-grade system-on-modules based on the same Qualcomm architecture without redesigning the complete software stack.

Berkeley Humanoid Lite cuts robot costs using modular actuators

Humanoid robots, including smaller-scale designs, can be expensive to develop and build. The Berkeley Humanoid Lite takes a low-cost approach by using 3D-printed actuators that form a significant part of the robot’s structure. The project is designed as an open source and customisable humanoid robot reference platform.

At the centre of the design are modular actuators that combine motors, 3D-printed cycloidal gearboxes and embedded magnetic encoders. The modular approach means that the actuators can potentially be used independently in other robotics projects, without requiring developers to reproduce the complete humanoid robot.



Rather than being presented as a finished commercial product, Berkeley Humanoid Lite is intended to serve as an accessible platform for experimentation and modification. The project’s GitHub repository contains the resources needed to build and customise the robot, while demonstrations include virtual reality (VR)-driven teleoperation.